

《概率论与数理统计》试卷（3 学分 A）参考答案

一. 选择题

A B A B (B, C) .

二. 填空题

1. $\frac{5}{7}$;

2.. 4 ;

3. $5/9$;

4. $7/8$ (或 0.875) ;

5. $(\bar{X} - \frac{S}{\sqrt{n}} t_{\frac{\alpha}{2}}(n-1), \bar{X} + \frac{S}{\sqrt{n}} t_{\frac{\alpha}{2}}(n-1))$;

6. 9, t.

三解：由全概率公式及 Bayes 公式

$$P(\text{该种子能发芽}) = 0.1 \times 0.9 + 0.9 \times 0.2 = 0.27$$

$$P(\text{该种子来自发芽率高的一盒}) = (0.1 \times 0.9) / 0.27 = 1/3$$

四. 解: (1) 当 $0 < x < 1$ 时 $f_X(x) = \int_x^1 6xdy = 6x(1-x)$ 故

$$f_X(x) = \begin{cases} 6x(1-x) & 0 < x < 1 \\ 0 & \text{其他} \end{cases}$$

$$\text{当 } 0 < y < 1 \text{ 时, } f_Y(y) = \int_0^y 6xdx = 3y^2 \quad \text{故 } f_Y(y) = \begin{cases} 3y^2 & 0 < y < 1 \\ 0 & \text{其他} \end{cases}$$

$$(2) \text{ 当 } \frac{1}{3} < y < 1 \text{ 时, } f_Y(y|X=\frac{1}{3}) = \frac{f(\frac{1}{3}, y)}{f_X(\frac{1}{3})} = \frac{3}{2},$$

$$\text{故 } f_Y(y|X=\frac{1}{3}) = \begin{cases} \frac{3}{2} & \frac{1}{3} < y < 1 \\ 0 & \text{其他} \end{cases}.$$

$$(3) \quad P(X+Y \leq 1) = \int_0^{1/2} 6xdx \int_x^{1-x} dy = \int_0^{1/2} 6x(1-2x)dx = \frac{1}{4}.$$

五、解: 由题意知 X, Y 相互独立, 且

$$f_X(x) = \begin{cases} 2e^{-2x} & x > 0 \\ 0 & x < 0 \end{cases} \quad \text{与} \quad f_Y(y) = \begin{cases} e^{-y} & y > 0 \\ 0 & y < 0 \end{cases}.$$

$$\begin{aligned} \text{当 } z > 0 \text{ 时,} \quad F_Z(z) &= P\{\max(X, Y) \leq z\} = P\{X \leq z, Y \leq z\} \\ &= P\{X \leq z\}P\{Y \leq z\} = F_X(z)F_Y(z) \end{aligned}$$

$$f_Z(z) = f_X(z)F_Y(z) + F_X(z)f_Y(z) = 2e^{-2z}(1 - e^{-z}) + (1 - e^{-2z})e^{-z} = e^{-z} + 2e^{-2z} - 3e^{-3z}$$

$$\text{故} \quad f_Z(z) = \begin{cases} e^{-z} + 2e^{-2z} - 3e^{-3z} & z > 0 \\ 0 & \text{其他} \end{cases}$$

$$\text{六 解: } X \text{ 的分布函数} \quad F(x) = \begin{cases} 1 - e^{-\frac{1}{20000}(x-365)}, & x \geq 365, \\ 0, & x < 365 \end{cases},$$

$$\text{于是} \quad P(X \leq 1095) = 1 - e^{-0.0365} \approx 0.04$$

$$\text{记} \quad \begin{cases} N = \text{"1000件产品中寿命小于1095的产品件数"} \\ Y = \text{"保险公司的利润"} \end{cases}$$

$$\text{则 } N \sim B(1000, 0.04), \quad Y = 1000 \times P_0 - 2000N,$$

$$\text{由中心极限定理, } N \sim N(40, 6.2^2),$$

于是

$$(1) \quad \text{若保费 } P_0 = 100 \text{ 元/件, 则 "保险公司亏本"} = \{Y \leq 0\} = \{N \geq 50\}$$

$$P\{\text{保险公司亏本}\} = P\{Y \leq 0\} = P\{N \geq 50\} = P\left\{\frac{N-40}{6.2} \geq \frac{10}{6.2}\right\} \approx 1 - \Phi(1.61) = 0.054$$

$$(2) \quad \text{若保费为 } P_0, \text{ 则 "保险公司亏本"} = \{Y \leq 0\} = \{N \geq 0.5P_0\}$$

$$P\{\text{保险公司亏本}\} = P\{N \geq 0.5P_0\} = P\left\{\frac{N-40}{6.2} \geq \frac{0.5P_0-40}{6.2}\right\} \approx 1 - \Phi\left(\frac{0.5P_0-40}{6.2}\right) \leq 0.01$$

$$\begin{aligned} \text{故} \quad \Phi\left(\frac{0.5P_0-40}{6.2}\right) &\geq 0.99 \Rightarrow \frac{0.5P_0-40}{6.2} \geq 2.33 \\ &\Rightarrow P_0 \geq 2 \times (40 + 6.2 \times 2.33) = 108.89 \text{ (元)} \end{aligned}$$

七解: (1) Y 的概率密度为:

$$f(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(y-\mu)^2}{2}}, -\infty < y < +\infty, \text{ 于是, (令 } t = y - \mu \text{)}$$

$$\begin{aligned}
 b = EX &= Ee^Y = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^y e^{-\frac{(y-\mu)^2}{2}} dy \\
 &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{t+\mu} e^{-\frac{t^2}{2}} dt \\
 &= e^{\mu+\frac{1}{2}} \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(t-1)^2} dt = e^{\mu+\frac{1}{2}}
 \end{aligned}$$

(2) 当置信度 $1-\alpha=0.95$ 时, $\alpha=0.05$ 。标准正态分布的水平为 $\alpha/2=0.025$ 的分位数为 $\mu_{0.025}=1.96$ 。故由 $\bar{Y} \sim N(\mu, 1/4)$, 可得

$$P\left\{\left|\frac{\bar{Y}-\mu}{1/2}\right| < 1.96\right\} = P\left\{\bar{Y} - \frac{1}{2} \times 1.96 < \mu < \bar{Y} + \frac{1}{2} \times 1.96\right\} = 0.95$$

其中

$$\bar{Y} = \frac{1}{4}(\ln 0.5 + \ln 0.8 + \ln 0.125 + \ln 2) = \frac{1}{4} \ln 1 = 0$$

于是, 有

$$P\{-0.98 < \mu < 0.98\} = 0.95$$

从而 $(-0.98, 0.98)$ 就是 μ 的置信度为 0.95 的置信区间。

八. 解: (1) $H_0: \mu=500$ $H_1: \mu \neq 500$.

若 H_0 成立, 统计量 $T = \frac{\bar{X}-500}{S/3} \sim t(8)$.

由备择假设知, 拒绝域的形式为 $D = \{|\frac{\bar{X}-500}{S/3}| > A\}$,

由 $P\{T \in D | H_0\} = P\{|T| > A | H_0\} = \alpha$ 知 $A = t_{0.025}(8) = 2.306$.

故拒绝域为 $D = \{|T| > 2.306\}$.

代入数据得 T 的观察值 $T_0 = -\frac{3}{16.03} = -0.187$, 因 $T_0 \notin D$, 故接受 H_0 .

(2) $H_0: \sigma^2 = 100, H_1: \sigma^2 \neq 100$

由备择假设 H_1 知, 拒绝域的形式为 $C = \{\frac{8S^2}{100} < B\} \cup D = \{\frac{8S^2}{100} > A\}$.

在 $H_0: \sigma^2 = 100$ 成立的情况下, $\chi^2 = \frac{8S^2}{\sigma^2} \sim \chi^2(8)$,

取 $A = \chi_{0.025}^2(8) = 17.5$, $B = \chi_{0.975}^2(8) = 2.18$

代入数据得 $\frac{8 \times 16.03^2}{100} = 20.56 \in D$, 故应拒绝 H_0 .

九. (7 分) 证明: 由题设条件知 $ABC \subset D \Rightarrow P(ABC) \leq P(D)$,

$$\begin{aligned} P(A) + P(B) - P(AB) &\leq 1 \\ \Rightarrow P(A) + P(B) &\leq 1 + P(AB) \\ \Rightarrow P(A) + P(B) + P(C) &\leq 1 + P(AB) + P(C) \\ &= 1 + P(AB \cup C) + P(ABC) \\ &\leq 2 + P(ABC) \\ &\leq 2 + P(D) \end{aligned}$$